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/ **Practice Exam - 1**

Correct Answer

Next Question

Practice Exam - 1

Question 61 of 117



A 73-year-old male with stage D ischemic cardiomyopathy, diabetes, hypertension and hyperlipidemia presents to VAD clinic for a routine visit 12 months after DT HVAD implantation.

He reports he is doing well with excellent functional status. He reports poor dietary habits over the prior 6 months and has noted a 15 pound weight gain. He otherwise reports excellent adherence with his medications and has been following with his local doctors routinely. He reports recently noticing a gradual decrease in his HVAD flows from 5-6 L/min previously to low 3s and has had 1 low flow alarm.

In the office today, examination is notable for a loud VAD hum, JVP estimated at 7 cm of water, no lower extremity edema and driveline appears clean, dry and intact. His blood pressure measured by Doppler is 118/87 mmHg. His HVAD waveforms are notable for a mean flows of ~ 2.8 L/min, peak flows of 6.5L/min and nadir flows of 1 L/min.

His blood work is unrevealing. TTE shows intermittent aortic valve opening, mild RV dysfunction, trivial tricuspid/mitral regurgitation and midline septum.

Which of the following is the best next step to manage this patient?

A. Decrease VAD speed



B. Increase VAD speed

C. Increase bumex dose



D. Initiate losartan therapy

Commentary

References

Testing Point: The Board-certified AHFTC specialist should recognize hypertension as a possible cause of low flow alarms in VAD supported patients.

Answer: D. Initiate losartan.

The patient is presenting to the VAD clinic with progressive decline in HVAD flows in the setting of significantly elevated blood pressure. His recent weight gain and dietary indiscretion are likely contributors to worsening of his blood pressure control. Elevated blood pressure increases afterload of the pump i.e. the head pressure seen by the pump as demonstrated by the HQ curves published for individual devices. Notably, the detrimental effect of hypertension on flow rates is more prominent during diastole and so peak flow rates can remain stable or see mild decreases while the effect on mean and trough flows is far more significant. Further, uncontrolled hypertension is a known risk factor for stroke - one of the most debilitating adverse events seen after VAD implantation. As such, initiation of antihypertensives is the first step in managing this patient.

Answer A is not the correct answer for several reasons. While decreasing VAD speed may decrease blood pressure, echocardiogram shows optimized VAD speed and he otherwise feels well. Decreasing VAD speed can further decrease flows and potentially increase symptoms. The correct approach to addressing his hypertension is the initiation of antihypertensives.

Answer B. The less knowledgeable AHFTC physician may select this option if they do not recognize that hypertension rather than inappropriate VAD speed is driving the low flows. Increasing VAD speed in this clinical scenario may increase flows slightly, however speed increases can also worsen the already poorly controlled blood pressure. This, in turn, results in only slight changes in flows at the expense of even further increasing the patient's risk for stroke and other detrimental effects of hypertension. If RV dysfunction is present, increasing the VAD speed may actually worsen the right-sided HF symptoms due to increased RV preload.

Answer C. Increasing diuretic dosage is not the correct answer as volume overload and RV dysfunction are not the primary drivers of the patient's low flows. RV dysfunction, which is exacerbated by fluid overload is a cause of underfilling of the LV and VAD and is on the differential of low flow alarms in a VAD supported patient. However, there is no evidence of RV failure contributing to the clinical scenario presented here. The patient JVP demonstrates euvoemia and the echocardiogram reveals only mild RV dysfunction, trivial TR and midline septum all arguing against RV dysfunction/failure causing low flow alarms. Further, in HVAD patients with RV failure causing low VAD flows, peak, mean and trough flows would all be depressed leading to low flows and low pulsatility. This patient is demonstrating low flows and high pulsatility consistent with a hypertensive waveform.

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